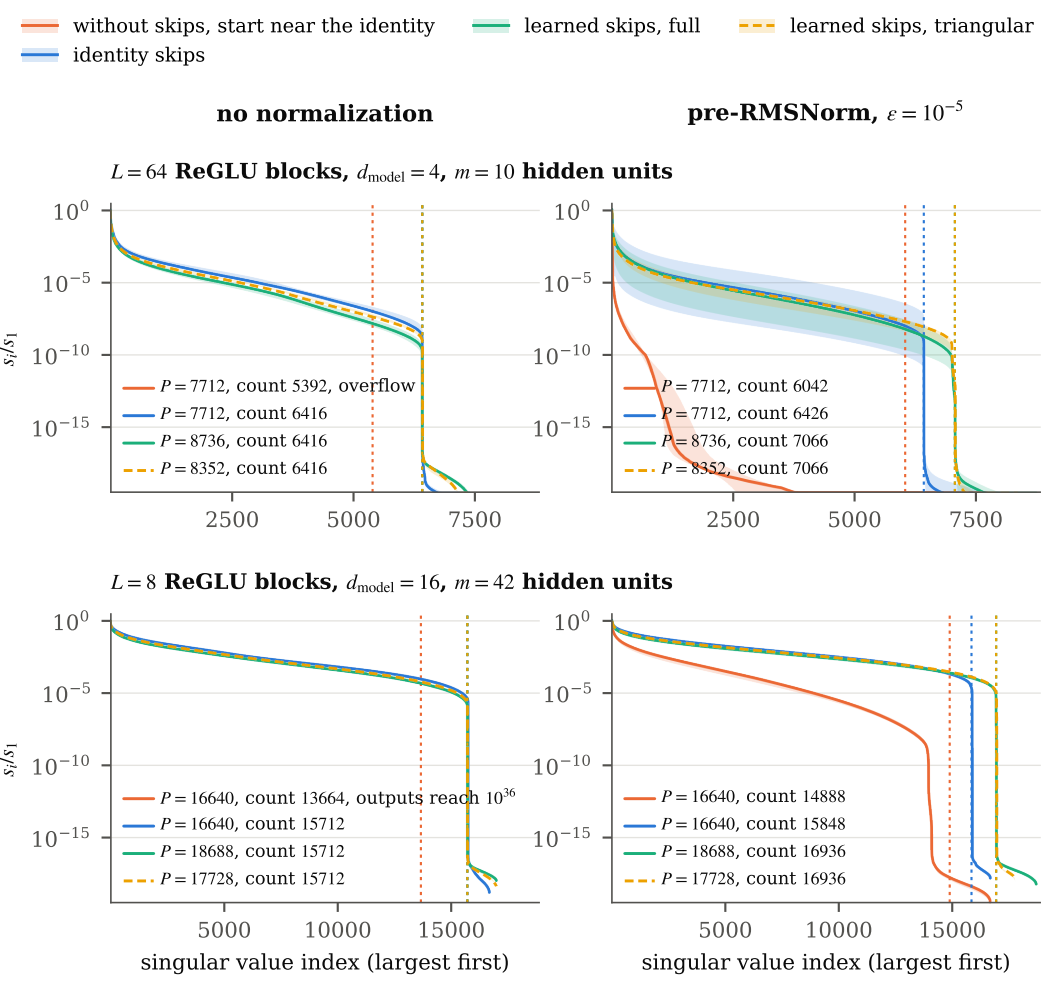

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ ReGLU blocks, $d_{\text{model}} = 4$, $m = 10$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 7712$, count 5392, overflow
$P = 7712$, count 6416
$P = 8736$, count 6416
$P = 8352$, count 6416
2500
5000
7500
$P = 7712$, count 6042
$P = 7712$, count 6426
$P = 8736$, count 7066
$P = 8352$, count 7066
$L = 8$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$P = 16640$, count 13664, outputs reach 10^{36}
$P = 16640$, count 15712
$P = 18688$, count 15712
$P = 17728$, count 15712
5000
10000
15000
singular value index (largest first)
$P = 16640$, count 14888
$P = 16640$, count 15848
$P = 18688$, count 16936
$P = 17728$, count 16936
5000
10000
15000
singular value index (largest first)